



POLITECNICO
MILANO 1863

2025–2026

Group 8

FINANCIAL ENGINEERING

Assignment n°4 - BS

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1 Introduction

This report presents a Statistical Arbitrage strategy applied to the Euro Stoxx 50 constituents, based on the Principal Component Analysis (PCA) framework proposed by Avellaneda and Lee (2008). The core methodology decomposes daily stock returns into systematic and idiosyncratic components, modeling the latter as a mean-reverting Ornstein-Uhlenbeck process. By standardizing these residuals into a dimensionless *s-score*, we generate systematic contrarian signals to exploit temporary price dislocations. In the following sections, we analyze the strategy's backtested performance—including the impact of transaction costs—and present a sensitivity analysis evaluating the robustness of different entry thresholds and dynamic targets for the variance explained by the principal components.

2 Implement the proposed PCA investment strategy

The Assignment asks us to implement a Principal Component Analysis using a one year lookback period and consider only the first four factors identified by the PCA for computing the linear multifactor model. Then, we use a 60 day lookback period and an AR(1) discrete model in order to estimate the univariate Ornstein-Uhlenbeck process for every residual series, in particular we do **not** de-mean the residual, we then select the stocks with mean-reversion speed greater than 8.4 (ie, we require an half-life less than 30 days) and compute the modified s-scores, including the drift term. It is also requested, in order to transform the signal into weights to use equally weighting strategy on all active signals and target a gross exposure of 100%. The results under the two scenarios requested in the analysis are presented in the following plot:

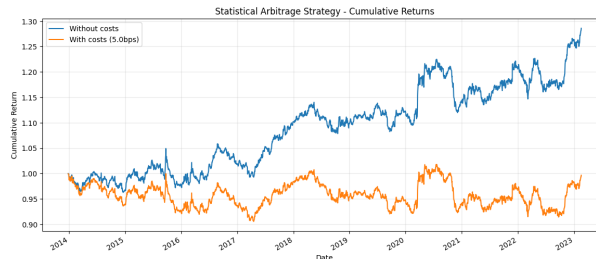


Figure 1: Statistical Arbitrage Strategy - Cumulative Returns

This plot highlights the impact of transaction costs on the cumulative returns of our strategy. In order to explain the variance, we used 4 principal components as requested by the Assignment, which led us to a total explained variance equal to 55.44%. This is a value close to the one discussed in Section 5.4 of Avellaneda and Lee (2008), where it is explained that targeting 55% explained variance yields a strong performing strategy. Using a smaller number of factors (4 in our case) leaves the residuals with a greater variance compared to a higher PCA threshold (e.g., 75%). While a larger residual variance implies larger deviations from equilibrium and theoretically wider arbitrage spreads, the plot clearly shows that these spreads are not wide enough to absorb the trading frictions. Statistical arbitrage strategies with daily rebalancing generate a very high portfolio turnover. When we apply an all-in transaction cost of 5 basis points, the expenses from frequent trading rapidly compound and completely cancel the gross profits, pushing the

net cumulative returns down to approximately 0.95. This demonstrates a crucial practical consideration: even if the 4-factor model generates good entry signals, the average profit per trade is too small to overcome the 5 bps transaction costs, completely changing the upward trend of the 'no cost' scenario into a losing strategy.

IMPORTANT note on the results just obtained:

As a group, we want to highlight the fact that this obtained result is quite a **warning signal** that something is wrong in the computation of the s-score and/or the *backtest.py* functions. We get it this as a warning signal because from the paper we expected to have, for a PCA equal to 55% ca, a much better cumulative return for the 'with costs' case. The expected case from our point of view, as specified in the previous lines of comments to the results, was to have a cumulative return for the 'with costs' case much closer to the one without costs, as we expected that the trading strategy, leaving much more room for spread over expected return, could absorb much better the costs. However, we reread the functions that we constructed to compute these results and we are in line with the formulas given in the paper from our point of view. Of course this kind of result highlights something strange that will extend also in the results that we got in the following points. We also really care to point out that the paper analyses the market in a different period from the one that is analyzed by us, so the result may be different from the one that we expected by the paper also because of this particular thing.

The discussion made in the previous point is also highlighted really much in the following plot:

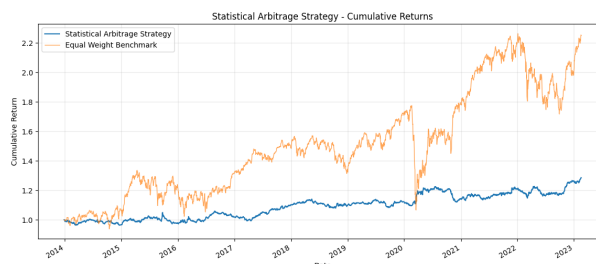


Figure 2: Statistical Arbitrage Strategy Vs 'Passive' Strategy

While the PCA-based statistical arbitrage is theoretically market-neutral and aims to generate absolute returns (alpha) uncorrelated with the market, the benchmark represents a naive, low-turnover passive allocation (beta). As highlighted in the plot, when accounting for the 5 bps transaction costs, the high turnover of our mean-reversion strategy completely erodes its profitability, causing it to vastly underperform the simple equal-weighted market drift.

3 Sensitivity analysis:

3.1 Robustness of the analysis against the opening threshold

The Assignment asks us to check the robustness of the strategy against the opening threshold. In order to do so, we study the strategy performance varying the value of the s-bo, ie threshold to open long position, between 1.0, 1.5, 2. Here below are presented the results:

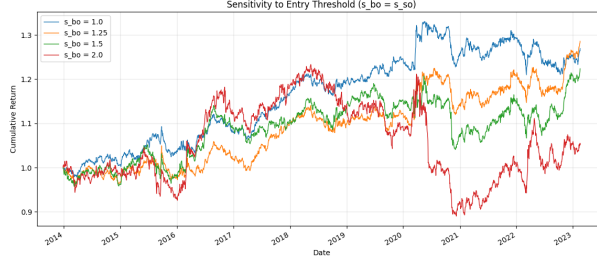


Figure 3: Performance by Entry Threshold

s_bo	Ann. Return	Ann. Vol	Sharpe	Max DD
1.00	2.62%	4.92%	0.53	-9.08%
1.25	2.74%	5.54%	0.49	-8.55%
1.50	2.20%	6.47%	0.34	-13.79%
2.00	0.56%	9.23%	0.06	-27.84%

Table 1: Performance by Entry Threshold

The strategy exhibits a clear and economically intuitive trade-off. Lowering the entry threshold from 2.00 to 1.00 increases the number of trading signals, allowing the portfolio to capture a larger fraction of the mean-reversion opportunities present in the residuals. This translates into higher total and annualised returns and a better Sharpe ratio at the 1.00 level. However, the increased signal frequency also raises turnover and exposes the strategy to larger adverse price moves before the position is opened, resulting in progressively higher maximum drawdowns. The base threshold of ± 1.25 strikes an excellent balance: it maintains strong profitability while keeping risk (volatility and drawdown) under control. Thresholds above 1.50 become excessively conservative, drastically reducing the number of active positions and leaving the portfolio vulnerable to prolonged periods of inactivity.

3.2 2. Sensitivity analysis: variable number of Principal Components per explained variance target

To evaluate the robustness of the strategy regarding the choice of the number of systematic factors, a backtest was implemented where, for each rebalancing date, the number of principal components (PCs) is determined dynamically to explain at least a predefined target of total variance: 40%, 55%, 65%, and 75%. The number of factors is calculated using the eigenvalues of the PCA estimated on a 252-day rolling window.

The obtained results are reported in the following table:

Table 2: Fixed vs Variable Factor Strategy Comparison

Strategy	Avg N. Factors	Total Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Fixed 4 factors	4	28.61%	2.74%	5.54%	0.49	-8.55%
Variable (40%)	1.4	32.16%	3.04%	6.80%	0.45	-21.69%
Variable (55%)	3.5	37.48%	3.48%	6.58%	0.53	-13.94%
Variable (65%)	6.8	38.69%	3.58%	5.62%	0.64	-8.11%
Variable (75%)	12	11.10%	1.14%	5.25%	0.22	-13.80%

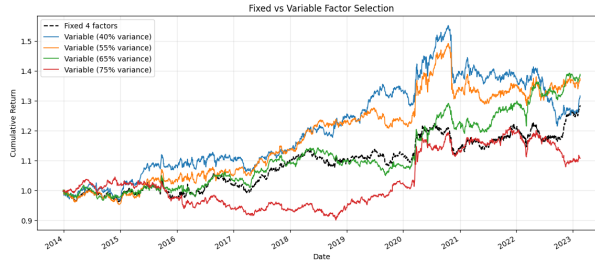


Figure 4: Fixed Vs Variable factor selection

The analysis reveals a significant sensitivity of the strategy to the explained variance parameter. In this specific dataset, the best risk-adjusted performance is achieved with the 65% target, which utilizes an average of approximately 7 factors. In this case, the strategy records a Sharpe Ratio of 0.64 and the lowest Max Drawdown (-8.11%).

As the target increases to 75%, a sharp deterioration in performance is observed, with the Total Return dropping to 11.10% and the Sharpe Ratio to 0.22. Conversely, the 40% target—while maintaining a positive return—exhibits much higher volatility (6.80%) and a more pronounced Max Drawdown (-21.69%).

3.3 Interpretation of results:

This trend is consistent with the market structure of the European index (SX5E). The first principal component typically explains between 43% and 62% of the total variance, making subsequent factors marginal. Using a reduced number of factors (40% target) leaves a larger idiosyncratic component in the residuals, which retains strong mean-reversion properties exploitable by the statistical arbitrage strategy based on the Ornstein-Uhlenbeck model. However, this also leaves the strategy more exposed to systematic shocks, as evidenced by the higher drawdown.

As the target increases, systematic risk is progressively removed, but part of the tradable signal present in the residuals is also eliminated. The significant decline observed at the 75% target suggests an excessive "over-cleaning" of the residuals, which makes the s-scores less informative and the strategy less profitable.

These results differ from the observations of Avellaneda & Lee (2008) regarding the US market, where a higher number of factors was often found to be advantageous. On the SX5E basket, an intermediate approach (around 65% of explained variance) appears to be preferable for balancing return and risk. This highlights the importance of calibrating the number of factors based on the specific universe and the concentration of systematic risk in the market under analysis.

4 Optional: Volume Adjusted Returns:

The final point of the Assignment asks us to refine the investment strategy using volume data referring to the strategy implemented in Question 2 (four constant factors and original thresholds). These are the results:

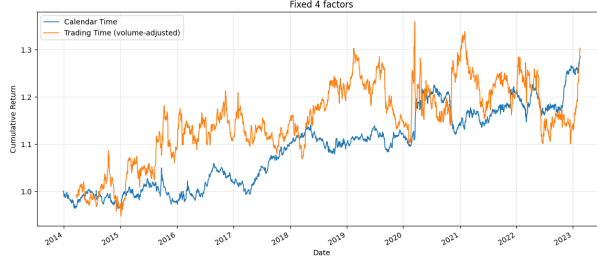


Figure 5: Strategy Comparisons

	Calendar Time	Trading Time
Total Return	28.61%	31.27%
Annualized Return	2.74%	3.04%
Annualized Volatility	5.54%	12.92%
Sharpe Ratio	0.49	0.24
Max Drawdown	-8.55%	-19.11%

Table 3: Strategy Comparison

The transition to Trading Time, following the volume-adjustment methodology of the paper, yields a marginal increase in total return (+2.66%) but results in a severe deterioration of risk-adjusted performance. Annualized volatility more than doubles, and the Maximum Drawdown expands significantly from -8.55% to -19.11%. Consequently, the Sharpe ratio collapses from 0.49 to 0.24, indicating that the refinement fails to improve the strategy's efficiency.

The economic rationale for this underperformance lies in the inverse correlation between liquidity and volatility during market stress. In periods like the March 2020 crash, trading volumes often dry up precisely when price moves are largest and most erratic. By normalizing returns through volume, the model amplifies noise and forces excessive exposure during illiquid regimes. While the authors found success with this adjustment in 2003–2007, our results suggest a regime shift where volume-based signals now introduce prohibitive tail risk. Therefore, the Calendar Time version remains the superior choice, offering a more stable and balanced risk-return profile.

5 Errors and notes in the code:

5.1 Ex4a-notebook:

Definition of parameters:

1. estimation_window was 2 years instead of 1 as requested by the Assignment.
2. n-factors: it was 5, but in the slide and Assignment was written 4
3. Buy to close threshold: it was 0.75, in the Assignment is actually 0.5
4. transaction costs: it was 10bps, it is actually 5bps

3. S-score and trading signals:

In the main the function s-score is taking as default, if not specified, that we want the modified s-score. However, in *statistical-arbitrage.py*, the function computes the **modified** s-score only if in the main you specify that you want to compute the s-score. This one is not really an error, one can also modify the default scenario directly in the function and not modify the main. However, we decided to underlie this fact.

5.2 statistical arbitrage.py:

estimate factor model: In this function the correlation matrix was computed as the variance-covariance matrix. This was wrong as it is written at the start of the function and in the paper. **compute s-score (important fact on how we handled the alphas):** In this function there were no errors, as it was requested to do it from scratch. The main thing that we want to underlie is how we handled the alphas that we need if we want to compute the modified s-scores. From our point of view, there were two main roads that we could follow:

1. Add an input to the *compute s-score* function, ie the alphas
2. Compute the alphas in the main and then "adding" them to *ou-parameters* in order to have them directly without modifying the main structure of the function.

Both the ways presented are correct from our point view. However, we decided to go with the second one in order to not modify the structure of the function and handling the problem in the function core and not from an input point of view. Although this is not part of the errors that we met in the code, we wanted to highlight this thing in order to give to the reader of the code an idea of what we were thinking during the structuring of the functions that we used to compute the s-scores.